

ENGR 213 · Applied Ordinary Differential Equations

Lecture 7: Non-Linear Mathematical Models (Explained)

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Advanced Engineering Mathematics

, 7th Edition, Section 2.8

Lecture Date: September 30, 2026

Executive Overview & Lecture Roadmap

Lecture 7 expands first-order mathematical modeling into **non-linear phenomena**. In Lecture 6, models were linear because the per-capita rate of change was assumed constant (e.g., $\frac{1}{P} \frac{dP}{dt} = k$). In real-world engineering, environmental limitations, resource depletion, molecular collisions, and fluid efflux create non-linear dependencies.

All three core physical models introduced in Lecture 7 lead to **separable first-order differential equations** solved using **partial fraction decomposition**:

Physical Phenomenon	Governing Non-Linear DE	Non-Linearity Mechanism	Key Physical Constant / Parameter
Population Dynamics	$\frac{dP}{dt} = P(a - bP) = \frac{aP - bP^2}{1}$	Fatal competition / crowding (quadratic $-bP^2$)	Carrying Capacity: $K = \frac{a}{b}$
Spread of an Epidemic	$\frac{dx}{dt} = kx(N - x)$	Interactions between infected (x) and uninfected ($N - x$)	Campus Population: N
Second-Order Chemical Reaction	$\frac{dX}{dt} = k(\alpha - X)(\beta - X)$	Collision frequency of reactants A and B	Limiting Reactant determines $\lim_{t \rightarrow \infty} X(t)$
Leaking Water Tank (Torricelli)	$\frac{dh}{dt} = -\frac{A_h}{A_w} \sqrt{2gh}$	Efflux speed scales with square root of head: $v = \sqrt{2gh}$	Emptying Time: $t_{\text{empty}} = \frac{A_w}{A_h} \sqrt{\frac{2H}{g}}$

1. Population Dynamics & The Logistic Equation

1.1 Physical Motivation: Why Exponential Growth Fails

In Lecture 6, the model for exponential growth assumed that the rate of population change is strictly proportional to the current population:

$$\frac{dP}{dt} = kP \implies P(t) = P_0 e^{kt} \quad (k > 0)$$

The **relative (or specific) growth rate** is defined as:

$$\frac{1}{P} \frac{dP}{dt} = k = \text{constant}$$

While accurate for short initial phases of bacteria or yeast cultures with unlimited nutrient supplies, **true exponential growth cannot continue indefinitely**. In any real physical or biological system, space, food, oxygen, and resources are finite. As the population expands, competition, waste accumulation, and resource depletion force the specific growth rate $\frac{1}{P} \frac{dP}{dt}$ to decrease.

This leads to the **density-dependent hypothesis**:

$$\frac{1}{P} \frac{dP}{dt} = f(P)$$

1.2 Derivation of the Logistic Model & Carrying Capacity

If an environment can sustain at most a fixed maximum population of K individuals, then K is called the **carrying capacity** of the environment. The growth rate must shut down when $P = K$:

1. $f(0) = r$ (maximum intrinsic birth rate when competition is zero).
2. $f(K) = 0$ (net growth ceases when the population reaches carrying capacity).

Assuming the simplest reasonable relationship—a **linear** decline in specific growth rate:

$$f(P) = c_1 P + c_2$$

Applying boundary conditions:

- $f(0) = c_2 = r$
- $f(K) = c_1 K + r = 0 \implies c_1 = -\frac{r}{K}$

Substituting $f(P)$ into the density-dependent equation:

$$\frac{1}{P} \frac{dP}{dt} = r - \frac{r}{K} P \implies \frac{dP}{dt} = P \left(r - \frac{r}{K} P \right)$$

Letting $a = r$ and $b = \frac{r}{K}$, we obtain the canonical **Verhulst Logistic Equation**:

$$\frac{dP}{dt} = P(a - bP) = aP - bP^2$$

- aP : The **vital growth term** (birth drive proportional to population).
- $-bP^2$: The **inhibition or competition term** (death/suppression rate driven by pairs of individuals competing for limited resources).
- **Carrying capacity**: $K = \frac{a}{b}$.

1.3 Analytical Solution via Partial Fractions

We solve the initial value problem:

$$\frac{dP}{dt} = P(a - bP), \quad P(0) = P_0 \quad (0 < P_0 < a/b)$$

Step 1: Separate Variables

$$\frac{dP}{P(a - bP)} = dt$$

Step 2: Partial Fraction Decomposition

Express the integrand as:

$$\frac{1}{P(a - bP)} = \frac{A}{P} + \frac{B}{a - bP}$$

Multiplying by $P(a - bP)$:

$$1 = A(a - bP) + BP$$

- Set $P = 0$: $1 = A(a) \implies A = \frac{1}{a}$
- Set $P = \frac{a}{b}$: $1 = B\left(\frac{a}{b}\right) \implies B = \frac{b}{a}$

Thus:

$$\frac{1}{a} \left(\frac{1}{P} + \frac{b}{a - bP} \right) dP = dt$$

Step 3: Integrate Both Sides

$$\int \left(\frac{1}{P} + \frac{b}{a - bP} \right) dP = \int a, dt$$

$$\ln |P| - \ln |a - bP| = at + C_1$$

$$\ln \left| \frac{P}{a - bP} \right| = at + C_1 \implies \frac{P}{a - bP} = Ce^{at}$$

Step 4: Apply Initial Condition $P(0) = P_0$

$$\frac{P_0}{a - bP_0} = Ce^0 = C$$

$$\frac{P}{a - bP} = \left(\frac{P_0}{a - bP_0} \right) e^{at}$$

Step 5: Solve Explicitly for $P(t)$

Multiply both sides by $(a - bP)$:

$$P = \left(\frac{P_0}{a - bP_0} e^{at} \right) (a - bP) = \frac{aP_0 e^{at}}{a - bP_0} - \frac{bP_0 e^{at}}{a - bP_0} P$$
$$P \left[1 + \frac{bP_0 e^{at}}{a - bP_0} \right] = \frac{aP_0 e^{at}}{a - bP_0}$$

Multiply the entire equation by $(a - bP_0)e^{-at}$:

$$P [(a - bP_0)e^{-at} + bP_0] = aP_0$$
$$P(t) = \frac{aP_0}{bP_0 + (a - bP_0)e^{-at}}$$

Dividing numerator and denominator by b :

$$P(t) = \frac{KP_0}{P_0 + (K - P_0)e^{-at}} \quad \text{where } K = \frac{a}{b}$$

1.4 Qualitative Behavior, Inflection Point & Asymptotes

- If an environment is capable of sustaining no more than a fixed number of K individuals in its population.
- The quantity **K** is called the **carrying capacity** of the environment, and:

$$\frac{dP}{dt} = f(P) \longrightarrow f(K) = 0 \quad \text{and} \quad f(0) = r \quad (\text{initial condition})$$

- The simplest assumption that we can make is that $f(P)$ is linear:

$$f(P) = c_1 P + c_2 \xrightarrow[\text{conditions}]{\text{applying the}} \frac{dP}{dt} = P \left(r - \frac{r}{K} P \right)$$

logistic equation

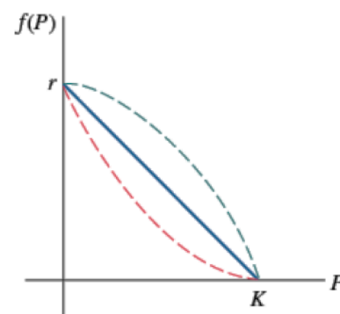


Figure 1: Logistic population growth curve, Lecture 7, Slide 4 (Zill Section 2.8).

Essential Qualitative Features:

1. Long-Term Asymptotic Limit:

$$\lim_{t \rightarrow \infty} P(t) = \lim_{t \rightarrow \infty} \frac{aP_0}{bP_0 + (a - bP_0)e^{-at}} = \frac{aP_0}{bP_0} = \frac{a}{b} = K$$

Regardless of whether the initial population starts small ($P_0 < K$) or large ($P_0 > K$), the population asymptotically converges to the carrying capacity K .

2. Point of Inflection (Maximum Growth Rate):

Differentiating the differential equation with respect to t using the chain rule:

$$\frac{d^2P}{dt^2} = \frac{d}{dt}[aP - bP^2] = (a - 2bP) \frac{dP}{dt}$$

Setting $\frac{d^2P}{dt^2} = 0$:

$$a - 2bP = 0 \implies P = \frac{a}{2b} = \frac{K}{2}$$

- **Physical Meaning:** The population grows fastest when it is **exactly at half its carrying capacity** ($P = K/2$).
- For $0 < P < K/2$: $\frac{d^2P}{dt^2} > 0$ (graph is concave up; accelerating growth).
- For $K/2 < P < K$: $\frac{d^2P}{dt^2} < 0$ (graph is concave down; growth decelerates as resources dwindle).

1.5 Lecture 7 — Example 1: Spread of a Flu Virus on Campus

- **Problem Statement (Slide 8):** Suppose a student carrying a flu virus returns to an isolated college campus of 1000 students. If it is assumed that the rate at which the virus spreads is proportional not only to the number x of infected students but also to the number of students not infected, determine the number of infected students after 6 days if it is further observed that after 4 days $x(4) = 50$.

Step 1: Formulate the Mathematical Model

- Let $x(t)$ = number of infected students at time t (days).
- Number of uninfected students at time $t = 1000 - x(t)$.
- Initial condition: $x(0) = 1$ (one carrier arrives).
- Benchmark condition: $x(4) = 50$.
- Governing differential equation:

$$\frac{dx}{dt} = kx(1000 - x)$$

Step 2: Separate Variables and Integrate

$$\frac{dx}{x(1000 - x)} = k, dt$$

Using partial fractions:

$$\frac{1}{1000} \left(\frac{1}{x} + \frac{1}{1000 - x} \right) dx = k, dt$$

$$\int \left(\frac{1}{x} + \frac{1}{1000 - x} \right) dx = \int 1000k, dt$$

$$\ln |x| - \ln |1000 - x| = 1000kt + C_1$$

$$\frac{x}{1000 - x} = Ce^{1000kt}$$

Step 3: Apply Initial Condition $x(0) = 1$

$$\frac{1}{1000 - 1} = Ce^0 \implies C = \frac{1}{999}$$

$$\frac{x}{1000 - x} = \frac{1}{999} e^{1000kt}$$

Step 4: Determine Rate Constant k Using $x(4) = 50$

$$\frac{50}{1000 - 50} = \frac{50}{950} = \frac{1}{19} = \frac{1}{999} e^{1000k(4)}$$

$$e^{4000k} = \frac{999}{19} \approx 52.5789$$

$$4000k = \ln \left(\frac{999}{19} \right) \approx 3.9623$$

$$1000k = \frac{3.9623}{4} \approx 0.9906 \text{ day}^{-1}$$

Step 5: Calculate Number of Infected Students at $t = 6$ Days

At $t = 6$:

$$1000k(6) = 6(0.9906) = 5.9435$$

$$\frac{x(6)}{1000 - x(6)} = \frac{1}{999} e^{5.9435} = \frac{1}{999} (381.25) \approx 0.3816$$

$$x(6) = 0.3816(1000 - x(6)) = 381.6 - 0.3816x(6)$$

$$1.3816x(6) = 381.6 \implies x(6) = \frac{381.6}{1.3816} \approx 276.2$$

- **Final Result:** After 6 days, approximately **276 students** are infected.
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2. Second-Order Chemical Reactions

2.1 The Law of Mass Action in Chemical Kinetics

Chemical kinetics investigates the rates at which chemical reactions proceed. Under the **Law of Mass Action**, the rate of a chemical reaction is proportional to the product of the active concentrations (or masses) of the reacting molecules.

- Suppose that chemicals A and B react and form chemical C.



* for this reaction, the rate at which the reaction proceeds is proportional to the product of the concentrations of CH_3Cl and of NaOH .

- Let's say, **a** grams of A are combined with **b** grams of B. If there are M parts of A and N parts of B in C, and **X(t)** is the mass of C formed, then the mass of A and B remaining at any time are:

$$a - \frac{M}{N+M}X \quad \text{and} \quad b - \frac{N}{N+M}X$$

- The rate of the reaction will be:

$$\frac{dX}{dt} \propto \left(a - \frac{M}{N+M}X\right)\left(b - \frac{N}{N+M}X\right) \longrightarrow \frac{dX}{dt} = k(\alpha - X)(\beta - X)$$

Figure 2: Second-order bimolecular chemical reaction formulation, Lecture 7, Slide 10 (Zill Section 2.8).

Derivation of the Reactant Mass Balance:

Consider two chemicals *A* and *B* combining to form a compound *C*:



- Suppose *M* parts of *A* combine with *N* parts of *B* to form *M + N* parts of *C*.
- Let *a* = initial mass of *A* (grams), and *b* = initial mass of *B* (grams).
- Let *X(t)* = mass of compound *C* formed at time *t*.

By stoichiometric conservation of mass:

- For each gram of *C* formed, $\frac{M}{M+N}$ grams of *A* are consumed.
- For each gram of *C* formed, $\frac{N}{M+N}$ grams of *B* are consumed.

Therefore, the masses of unreacted *A* and *B* remaining in the solution at time *t* are:

$$\text{Mass of } A \text{ remaining} = a - \frac{M}{M+N}X$$

$$\text{Mass of } B \text{ remaining} = b - \frac{N}{M+N}X$$

The rate of formation of C is proportional to the product of remaining reactants:

$$\frac{dX}{dt} \propto \left(a - \frac{M}{M+N}X\right) \left(b - \frac{N}{M+N}X\right)$$

Factoring out the stoichiometric fractions:

$$\frac{dX}{dt} = k_1 \left(\frac{M}{M+N}\right) \left(\frac{N}{M+N}\right) \left(a \frac{M+N}{M} - X\right) \left(b \frac{M+N}{N} - X\right)$$

Defining effective stoichiometric limits:

$$\alpha = a \left(\frac{M+N}{M}\right), \quad \beta = b \left(\frac{M+N}{N}\right), \quad k = k_1 \frac{MN}{(M+N)^2}$$

We arrive at the standard **Second-Order Rate Equation**:

$$\frac{dX}{dt} = k(\alpha - X)(\beta - X), \quad X(0) = 0$$

2.2 Analytical Integration

Case 1: Distinct Proportions ($\alpha \neq \beta$)

$$\frac{dX}{(\alpha - X)(\beta - X)} = k, dt$$

Decomposing by partial fractions:

$$\frac{1}{(\alpha - X)(\beta - X)} = \frac{1}{\alpha - \beta} \left(\frac{1}{\beta - X} - \frac{1}{\alpha - X} \right)$$

Integrating:

$$\begin{aligned} \frac{1}{\alpha - \beta} [-\ln |\beta - X| + \ln |\alpha - X|] &= kt + C_1 \\ \ln \left| \frac{\alpha - X}{\beta - X} \right| &= (\alpha - \beta)kt + C_2 \implies \frac{\alpha - X}{\beta - X} = C e^{(\alpha - \beta)kt} \end{aligned}$$

Applying initial condition $X(0) = 0$:

$$\begin{aligned} C &= \frac{\alpha}{\beta} \\ \frac{\alpha - X}{\beta - X} &= \frac{\alpha}{\beta} e^{(\alpha - \beta)kt} \end{aligned}$$

Asymptotic Behavior ($t \rightarrow \infty$):

If $\alpha > \beta$, then $(\alpha - \beta) > 0$. As $t \rightarrow \infty$, $e^{(\alpha-\beta)kt} \rightarrow \infty$, which forces:

$$\beta - X \rightarrow 0 \implies \lim_{t \rightarrow \infty} X(t) = \beta$$

- **Physical Principle:** The reaction terminates when the **limiting reactant** is completely exhausted. The maximum amount of C that can ever form is $\min(\alpha, \beta)$.

2.3 Lecture 7 — Example 2: Formation of Compound C

- **Problem Statement (Slide 12):** A compound C is formed when two chemicals A and B are combined. During the reaction, for each gram of A , 4 grams of B are used. It is observed that 30 grams of C are formed in 10 minutes. Initially there are 50 grams of A and 32 grams of B .
 - (a) Determine the amount of C at time t if the rate of reaction is proportional to the amounts of A and B remaining.
 - (b) How much of the compound C is present at 15 minutes?
 - (c) Interpret the solution as $t \rightarrow \infty$.

Step 1: Establish Stoichiometric Coefficients & Limits

- $M = 1$ part A , $N = 4$ parts $B \implies M + N = 5$ parts C .
- Fractions: Each gram of C consumes $\frac{1}{5}$ g of A and $\frac{4}{5}$ g of B .
- Initial amounts: $a = 50$ g of A , $b = 32$ g of B .
- Unreacted masses:

$$\text{Remaining } A = 50 - \frac{1}{5}X = \frac{1}{5}(250 - X)$$

$$\text{Remaining } B = 32 - \frac{4}{5}X = \frac{4}{5}(40 - X)$$

- Thus: $\alpha = 250$ and $\beta = 40$.
- The differential equation becomes:

$$\frac{dX}{dt} = k_1 \left(\frac{1}{5} \right) (250 - X) \left(\frac{4}{5} \right) (40 - X) = k(250 - X)(40 - X)$$

Step 2: Separate Variables & Integrate

$$\frac{dX}{(250 - X)(40 - X)} = k, dt$$

Partial fractions with $\alpha - \beta = 250 - 40 = 210$:

$$\frac{1}{210} \left(\frac{1}{40 - X} - \frac{1}{250 - X} \right) dX = k, dt$$

Integrating both sides:

$$\ln \left(\frac{250 - X}{40 - X} \right) = 210kt + C_1 \implies \frac{250 - X}{40 - X} = Ce^{210kt}$$

Step 3: Apply Initial Condition $X(0) = 0$

$$\frac{250 - 0}{40 - 0} = C \implies C = \frac{250}{40} = \frac{25}{4} = 6.25$$

$$\frac{250 - X}{40 - X} = \frac{25}{4}e^{210kt}$$

Step 4: Determine Rate Parameter Using $X(10) = 30$ g

At $t = 10$:

$$\frac{250 - 30}{40 - 30} = \frac{220}{10} = 22$$

$$22 = \frac{25}{4}e^{210k(10)} \implies e^{2100k} = \frac{22 \times 4}{25} = \frac{88}{25} = 3.52$$

$$2100k = \ln(3.52) \approx 1.25846$$

$$210k = \frac{1.25846}{10} = 0.12585 \text{ min}^{-1}$$

Step 5: Solve Explicitly for $X(t)$

$$\frac{250 - X}{40 - X} = \frac{25}{4}e^{0.12585t}$$

$$4(250 - X) = 25(40 - X)e^{0.12585t}$$

$$1000 - 4X = 1000e^{0.12585t} - 25Xe^{0.12585t}$$

$$X(t) [25e^{0.12585t} - 4] = 1000(e^{0.12585t} - 1)$$

$$X(t) = \frac{1000(e^{0.12585t} - 1)}{25e^{0.12585t} - 4} = \frac{1000(1 - e^{-0.12585t})}{25 - 4e^{-0.12585t}}$$

Part (b): Mass of C Present at $t = 15$ minutes

At $t = 15$:

$$0.12585 \times 15 = 1.88775$$

$$e^{1.88775} \approx 6.6045$$

$$X(15) = \frac{1000(6.6045 - 1)}{25(6.6045) - 4} = \frac{1000(5.6045)}{165.11 - 4} = \frac{5604.5}{161.11} \approx 34.787 \text{ grams}$$

- **Result:** At 15 minutes, approximately 34.79 g of compound C has formed.

Part (c): Asymptotic Interpretation as $t \rightarrow \infty$

$$\lim_{t \rightarrow \infty} X(t) = \lim_{t \rightarrow \infty} \frac{1000(1 - e^{-0.12585t})}{25 - 4e^{-0.12585t}} = \frac{1000(1 - 0)}{25 - 0} = \frac{1000}{25} = 40 \text{ grams}$$

- **Physical Interpretation:**

- Chemical B is the **limiting reactant**.
- Initially there were 32 g of B . Since each gram of C requires 0.8 g of B , the 32 g of B can produce at most:

$$X_{\max} = \frac{32 \text{ g}}{0.8 \text{ g of } B/\text{g of } C} = 40 \text{ grams of } C$$

- When $X = 40$ g, all 32 g of B is completely consumed.
- Mass of A consumed $= \frac{1}{5}(40) = 8$ g.
- Mass of A remaining $= 50 - 8 = 42$ g of unreacted A left in the container.

3. Draining / Leaking Tanks & Torricelli's Law

3.1 Fluid Mechanics Principles & Efflux Velocity

Consider a water tank filled to depth h draining through a small orifice of area A_h at the bottom.

- The speed v of efflux of water through a hole at the bottom of a tank filled to a depth h is: $v = \sqrt{2gh}$
- Imagine a water tank is draining through a hole with area of A_h , and we are interested in depth h of water remaining in the tank at time t .
- The volume of water leaving the tank per second is: $A_h \sqrt{2gh}$
- If $V(t)$ denotes the volume of water in the tank at time t : $\frac{dV}{dt} = -A_h \sqrt{2gh}$
- If the tank is such that volume of water in it at time t can be written $V(t) = A_w h$ (A_w is constant area of the upper surface of the water):

$$\frac{dV}{dt} = -A_h \sqrt{2gh} \xrightarrow{\frac{dV}{dt} = A_w \frac{dh}{dt}} \boxed{\frac{dh}{dt} = -\frac{A_h}{A_w} \sqrt{2gh}}$$

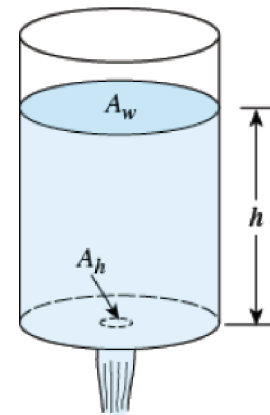


Figure 3: Efflux from a leaking water tank, Lecture 7, Slide 15 (Zill Section 2.8).

Torricelli's Law:

Under ideal (frictionless) conditions, equating potential energy at the free surface to kinetic energy at the orifice gives the **efflux velocity**:

$$v = \sqrt{2gh}$$

- h : Instantaneous depth of fluid above the orifice (*extm* or *extft*).
- g : Acceleration due to gravity (9.8 m/s^2 or 32 ft/s^2).

Volumetric Flow Rate Conservation:

The rate at which volume escapes through the hole is:

$$\frac{dV}{dt} = -A_h v = -A_h \sqrt{2gh}$$

If the cross-sectional area of the water's upper surface is $A_w(h)$, then $dV = A_w(h)dh$. Therefore:

$$A_w(h) \frac{dh}{dt} = -A_h \sqrt{2gh}$$

For a cylindrical tank with **constant surface area** A_w :

$$\frac{dh}{dt} = -\frac{A_h}{A_w} \sqrt{2gh}$$

3.2 Analytical Solution & Total Draining Time

We solve the initial value problem:

$$\frac{dh}{dt} = -\frac{A_h}{A_w} \sqrt{2g}, h^{1/2}, \quad h(0) = H$$

Step 1: Separate Variables

$$h^{-1/2} dh = -\frac{A_h}{A_w} \sqrt{2g} dt$$

Step 2: Integrate

$$2\sqrt{h} = -\frac{A_h}{A_w} \sqrt{2g} t + C$$

At $t = 0, h = H$:

$$C = 2\sqrt{H}$$

$$2\sqrt{h(t)} = 2\sqrt{H} - \frac{A_h}{A_w}\sqrt{2g}, t$$

$$\sqrt{h(t)} = \sqrt{H} - \frac{A_h}{2A_w}\sqrt{2g}, t$$

Step 3: Explicit Height Function

$$h(t) = \left(\sqrt{H} - \frac{A_h}{2A_w}\sqrt{2g}, t \right)^2$$

Step 4: Emptying Time (t_{empty}) & Interval of Definition

The tank is empty when water depth reaches zero ($h(t) = 0$):

$$\sqrt{H} - \frac{A_h}{2A_w}\sqrt{2g}, t_{\text{empty}} = 0$$

$$t_{\text{empty}} = \frac{2A_w\sqrt{H}}{A_h\sqrt{2g}} = \frac{A_w}{A_h}\sqrt{\frac{2H}{g}}$$

- **Crucial Physical Note on Interval of Definition:** The mathematical solution is a parabola $h(t) = (c_1 - c_2 t)^2$. However, after t_{empty} , the parabola curves back upward! In physical reality, water cannot spontaneously flow backwards out of an empty hole. Therefore, the **valid physical interval of definition** is strictly:

$$I = [0, t_{\text{empty}}]$$

For all $t > t_{\text{empty}}$, $h(t) \equiv 0$.

3.3 Lecture 7 — Example 3: Leaking Cylindrical Tank

- **Problem Statement (Slide 17):** A tank in the form of a right-circular cylinder standing on end is leaking water through a circular hole in its bottom.
 - (a) Solve for $h(t)$ if the initial height of water is H . Draw graph of $h(t)$ and give its interval I of definition in terms of the symbols A_w , A_h , and H .
 - (b) Suppose the tank is 10 ft high and has radius 2 ft and the circular hole has radius 0.5 in. If the tank is initially full, how long will it take to empty?

Part (a): General Formula & Interval of Definition

From our derivation above:

$$h(t) = \left(\sqrt{H} - \frac{A_h}{2A_w}\sqrt{2g}, t \right)^2$$

The tank completely empties at:

$$t_{\text{empty}} = \frac{A_w}{A_h} \sqrt{\frac{2H}{g}}$$

- **Interval of Definition:**

$$I = \left[0, \frac{A_w}{A_h} \sqrt{\frac{2H}{g}} \right]$$

Part (b): Numerical Calculation

- **Tank Height:** $H = 10 \text{ ft}$
- **Tank Radius:** $R = 2 \text{ ft}$
- **Hole Radius:** $r_h = 0.5 \text{ in} = \frac{0.5}{12} \text{ ft} = \frac{1}{24} \text{ ft}$
- **Gravitational Constant:** $g = 32 \text{ ft/s}^2$

Step 1: Compute Cross-Sectional Areas

$$A_w = \pi R^2 = \pi (2 \text{ ft})^2 = 4\pi \text{ ft}^2$$

$$A_h = \pi r_h^2 = \pi \left(\frac{1}{24} \text{ ft} \right)^2 = \frac{\pi}{576} \text{ ft}^2$$

Step 2: Compute Area Ratio

$$\frac{A_w}{A_h} = \frac{4\pi}{\pi/576} = 4 \times 576 = 2304$$

Step 3: Compute Emptying Time

$$t_{\text{empty}} = \frac{A_w}{A_h} \sqrt{\frac{2H}{g}} = 2304 \sqrt{\frac{2(10 \text{ ft})}{32 \text{ ft/s}^2}} = 2304 \sqrt{\frac{20}{32}} = 2304 \sqrt{0.625}$$

$$\sqrt{0.625} \approx 0.790569$$

$$t_{\text{empty}} = 2304 \times 0.790569 \approx 1821.47 \text{ seconds}$$

Step 4: Convert to Minutes

$$t_{\text{empty}} = \frac{1821.47}{60} \approx 30.36 \text{ minutes} \approx 30 \text{ min } 22 \text{ s}$$

4. First-Principles Exam Pitfalls & Non-Linear Traps

Trap 1: Dimensional Inconsistency in Torricelli's Law

- **The Error:** Mixing inches and feet ($R = 2 \text{ ft}$ with $r_h = 0.5 \text{ in}$).

- **The Rule:** Always convert all linear dimensions into feet before calculating areas:

$$r_h = 0.5 \text{ in} = \frac{1}{24} \text{ ft} \implies A_h = \frac{\pi}{576} \text{ ft}^2$$

Similarly, ensure g is in ft/s^2 ($g = 32$) for Imperial units, or m/s^2 ($g = 9.8$) for SI units.

Trap 2: Extending $h(t)$ Past the Emptying Time

- **The Error:** Plugging a time $t > t_{\text{empty}}$ into $h(t) = (\sqrt{H} - ct)^2$.
- **The Rule:** Because the square makes the expression positive, math says depth rises after emptying! In an engineering exam, state explicitly:

$$h(t) = \begin{cases} \left(\sqrt{H} - \frac{A_h}{2A_w} \sqrt{2g} t \right)^2, & 0 \leq t \leq t_{\text{empty}} \\ 0, & t > t_{\text{empty}} \end{cases}$$

Trap 3: Chemical Reaction Stoichiometric Fractions

- **The Error:** Writing $\frac{dX}{dt} = k(a - X)(b - X)$ when the reactants do not combine in a 1 : 1 mass ratio.
- **The Rule:** If M grams of A react with N grams of B , the consumption factors are $\frac{M}{M+N}$ and $\frac{N}{M+N}$. Factoring them out scales the effective limits:

$$\alpha = a \left(\frac{M+N}{M} \right), \quad \beta = b \left(\frac{M+N}{N} \right)$$

Trap 4: Logistic Inflection Point vs. Maximum Population

- **The Error:** Confusing the point of maximum growth rate with maximum carrying capacity.
- **The Rule:**
 - The **maximum growth rate** occurs at the inflection point: $P = \frac{K}{2}$.
 - The **maximum population** is the asymptotic limit: $P = K$.